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"Ethics against Chemistry": Solving a Crime Using Chemistry Concepts and Storytelling in a History of Science-Based Interactive Game for Middle School Students

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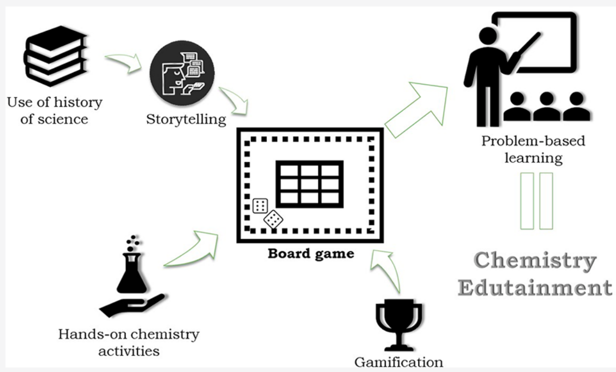
ABSTRACT: Designing a science activity for middle school children is a challenging task, especially if it aims to be interdisciplinary. One may ask if it is possible to craft a positive learning experience from different areas such as history of science, chemistry, or ethics. In this paper, we argue it can be achieved if we use the right tools to engage a young audience. A combination of gamification, hands-on activities, and storytelling techniques was successfully used, providing a unique form to approach chemistry and ethics subjects in an "edutainment" format. The result is a game that challenges participants to solve a crime, while addressing chemistry topics to gather pieces of evidence and facing science-related ethical conundrums. In our pilot study, participants evaluated this activity positively, identifying the innovation and entertaining features as the most relevant in their gaming experience.



KEYWORDS: *Elementary/Middle School Science, History/Philosophy, Interdisciplinary/Multidisciplinary, Hands-On Learning/Manipulatives, Problem Solving/Decision Making, Ethics*

■ INTRODUCTION

We have seen an increasing interest in using storytelling techniques for creating new science dissemination content.[1−3] Storytelling can be a valuable tool to raise students' perception and interest in science, especially in middle schoolers. It can be very useful to address controversial concepts such as moral issues to a younger audience,[4] while raising their awareness on the role of science in their lives.[5,6]

Hands-on activities are also an excellent resource to engage middle school students. They gain control of their learning process, increasing their motivation, interest, and curiosity of a specific topic.[2,7]

One of the most adopted strategies as an educational technique is gamification.[8,9] This approach is the application of game logic to a nongame context,[10] allowing students to play a game while they enjoy learning about subjects they may find tedious (or difficult) to understand. The inherent competitive factor serves as a learning enabler, aside from developing social, cognitive, and emotional skills.[11,12] Gamification is also helpful to introduce themes not usually discussed in the classroom[12] and has proven to be valuable for students to gain certain skills when simulating real situations.[13,14]

Different games (such as board games) are acknowledged as an effective teaching approach,[15−17] even in the exact sciences topics with a laboratory component.[18,19] Game-based logic activities and role-playing interaction such as "escape rooms", "crime laboratories", and "science quests" highlight the potential of using "real-life" situations as novel teaching foundations.[20−23] All of these examples embody a specific type of entertainment that has been considered to have an important and positive impact on science communication perception. By considering the emotional side of the audience, while educating and entertaining, we can name this as "edutainment".[24,25]

However, it takes more than an appealing game design to effectively communicate knowledge and scientific skills.[26,27] An excellent form of "concept transmission" is key when designing a game. To do so, it is imperative to engage players, encouraging them to solve tasks within a problem-based learning scenario. In those tasks, they transmit concepts through questions, ideas, and observations, allowing students to draw their own conclusions about emerging problems.[28] This model can be used to incorporate science concepts into the game, easily achieved through appealing storytelling.[29,30]

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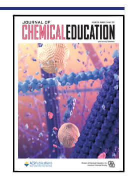

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The combination of "real-life-based" storytelling with gamification in a science communication activity, specifically in a multidisciplinary context, has high potential.

SCIENCE AND ETHICS TOPICS FOR MIDDLE SCHOOLERS

Now more than ever, our challenge as scientists, teachers, and science communicators is to engage audiences with contemporary educational issues, such as the limits of science and ethics.[31,32] Ethics can be defined as a set of moral principles that can give guidance on our conduct.[33] Nonetheless, it can be tough to discuss these topics in the classroom when the boundary between science and ethics is not well-defined. Areas like climate change, artificial intelligence, data protection, and medical advances hold significant ethical issues that must not be ignored. For chemistry, the synthesis of new products, either used as weapons or to improve the medical or agricultural fields, provides significant ethical tensions between chemistry itself and its application.[34]

The role of science ethics in education is well expressed in the literature,[35] stating that it can "bridge the gap between research and its ethical implications". It "can cover the relations between research and society", as some issues can be addressed in the classroom, easing student's perspective on science comprehension.[36] But what strategies can we use? A child can certainly understand the difference between right and wrong, with a simple ethics explanation. In the same sense, we can say that the "right" thing for science is to analyze facts while not being influenced by the scientist's convictions or personal beliefs.[31] As such, ethics can help underline that the scientific endeavor develops through fact-checking, trying to separate any personal situations of the scientists from the facts.

However, we must not expect that young students can easily grasp the ethical conundrums of the limits of science, but we should not ignore these topics altogether. For instance, the issue of "conflicts of interest in science", although not usually discussed with students, can be a good ethical starting point to bring into the classroom. To establish as illegitimate when someone has their own personal interests, undermining their professional activity, is not so farfetched to a middle schooler. The personification of a scientist dealing with scientific facts (or a police officer, politician, or any law-abiding citizen) is easily viewed by students like someone who acts in pursuing evidence and truth. As such, when someone acts on behalf of individual financial, personal, or professional gain, it can be explained as cases when a scientist (for example) must be sanctioned as an unethical discharge of his duties. This makes the case for approaching several real-life situations that help illustrate such an event.

Moreover, this also highlights the importance of bringing these issues to a middle school science environment more than a university science environment. By introducing concepts of moral dilemmas in science activities, we can easily make the bridge for the present issues of science facts being produced with misinformation toward a general audience. The creation of fake-scientific evidence (or "fake news") or tampering with an investigation are topics thoroughly explained in young people's media, such as videogames, television shows, and other digital entertainment content. Introducing these issues to such a young audience should raise student awareness, as they grow to be active members of their community.

HISTORY OF SCIENCE IN THE CLASSROOM

The rich and diverse stories throughout the development of the scientific endeavor can be used as templates to highlight specific topics.[37,38] For example, key achievements in forensic science[38] can be found in several stories that intertwine traits of crime episodes. Previous workshops of forensic science have shown real-world scenarios to be a good teaching tool.[39,40]

Recent reviews explore how the history of science can have an important part in the science classroom.[41] History of science can help teachers approach students' science learning difficulties through exemplification of scientists' struggles to comprehend chemistry reactions and how sometimes these individuals were despised, despite having a newfound view on a subject.[42] More importantly, students can relate with these eminent scientists, as even they struggled to understand science itself. As such, student attitude toward a science learning environment can be improved. History of science episodes can show how the discovery of an alternative theory or novel substance might be done by any common citizen.[43] For the case of this article, one episode of the history of science is viewed as a thoughtful supplier of narratives that teachers (and science communicators) can use. We do not intend to spark a debate on historical perspectives but rather to expose that these real-life situations can be used to familiarize students with a specific learning environment.

In this activity, guided by a teacher, students will have to solve problems and come up with their conclusions, while learning at the same time. They will achieve conceptual development by identifying problems, setting the scientific method, conducting experiments, and interpreting data. This promotes the operationalization of scientific knowledge, which is shown and explained in their own rhythm.

ACTIVITY FRAMEWORK

"Ethics against Chemistry"[44] is an educational science dissemination chemistry course, integrated into the 2019 Junior University program of activities of the Faculty of Sciences of the University of Porto (Portugal). This summer program aims to introduce children to a university environment.[45] It was presented to middle school students under 14 years old as a series of hands-on activities, along with gamification elements based on scientific events. This distinctive approach centered on storytelling, inspired by facts and figures of a Portuguese 19th century forensic murder case. It all comes together in a board game, composed of "stations", challenging the students to fulfill a game report to complete the game.

As part of the game's tasks, students are requested to perform experimental activities to solve and answer the main game enigma: Who committed the murder? To succeed, students team up and play the role of scientists/investigators, following the scientific method, and solve this mystery. They perform actions such as analysis of a crime scene, handling and treatment of samples, and executing scientific procedures based on methodical decisions. They are also faced with conflict-of-interest challenges based on the real story. As a requirement to complete the game, students exhibit their findings in a conference-like presentation and are encouraged to argue the scientific and ethical implications of the procedures used.

In Figure 1, we outline a schematic representation of "Ethics against Chemistry". To faithfully adapt this activity within the contents addressed by students in their seventh/eighth grade

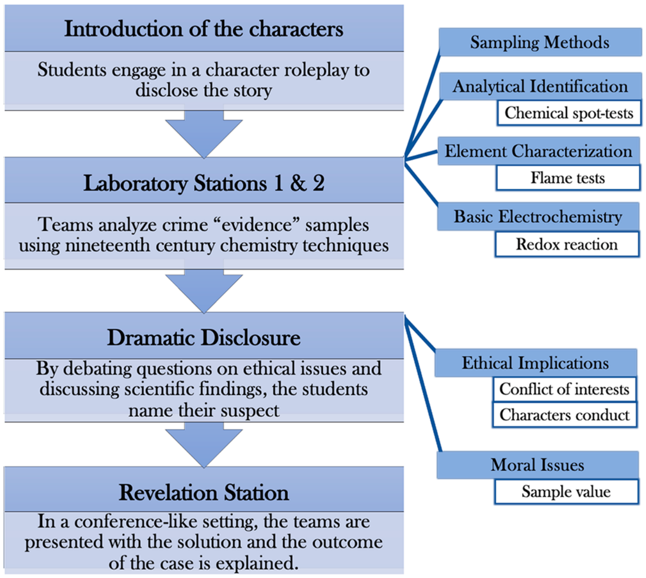

Figure 1. Schematic summary of the structure and goals of the activity "Ethics against Chemistry".

Chemistry class, we incorporated "Curricular Goals" on the learning sections of the activity.[46] For the four items discussed in the laboratory stations, we used the following goals.[47,48]

For the seventh grade students, we addressed these three specific curricular goals in "Ethics against Chemistry":

1. Recognize precipitation reactions with the formation of poorly water-soluble salts.
2. Differentiate the meaning of "pure" substance in everyday applications and in chemistry contexts.
3. Classify a mixture by its macroscopic aspect as heterogeneous or homogeneous.

For the eighth grade students, we addressed these five specific curricular goals in "Ethics against Chemistry":

1. Associate material corrosion to a type of chemical reaction as an oxidation−reduction reaction.
2. Give examples of acid, basic, and neutral aqueous solutions used in the laboratory and at home.
3. Classify aqueous solutions as acid, basic, or neutral, based on the behavior of colorimetric indicators.
4. Understand that catalysts and inhibitors are not consumed in chemical reactions but can lose their activity.
5. Interpret the variation of reaction speed, based on the control of the factors that change it.

The storytelling elements feature a roleplaying dynamic, providing an interesting learning game environment. Students, as investigators, will meet all the characters of the story: suspects, victims, police officers, and science experts.

A digital presentation sets the roleplaying dynamic of the game. The students give voice to the characters, interacting with each other. This presentation serves as a time frame for the board game, instructing teams to move their pins across the board's stations. When they reach the laboratory stations, the story pauses so that they can perform the planned experiments. The story resumes in the same framework, leading to the final game station, the scientific conference. In this part, students elaborate on their findings and reveal who is their suspect, based on the evidence analyzed and story context. Throughout the activity, they will compose a full scientific report using all the information available, as digital and Internet-based searches (or other external inputs) are not permitted.

At the end of each station, students are asked to share their suspicions on their game report. Every character in the story can be an eligible suspect (except the ones who die). This allows teachers to track students' thoughts, evaluating if their scientific findings (or if the final ethical dilemma) had an impact on their answers. To win the game, teams have to present the correct final answer (who was the murderer) in a coherent and valid scientific report. In the case of a tie (or in the event any team is incapable of identifying the culprit), intermediate answers are analyzed, and the team that presents more accurate findings, by comparison, is declared the winner. A prize is awarded to the victorious team.

In addition, some story elements of storytelling are purposefully misleading. However, they do not taint in any way the scientific findings of the laboratory stations and the outcome of the game.

■ GOALS

This activity encompasses learning and pedagogical goals. Evaluation of learning outcomes is done though questionnaires and reports (presented in Supporting Information). The intended goals are

- Use a real historic case as a base to teach chemistry in a fun way
- Evaluate students' scientific weighing of an ethical dilemma
- Evaluate the level of acceptance of participants

At this pilot stage, evaluation of the accomplishment of these objectives is simple (adapted to a sheer exploratory status). However, we will characterize them through our results, based on the evaluation performed by students on the activity. This will be done in a semiquantitative way, complementing a numerical score with a small description. Facing a relatively small sample, this is to be comprehended as a valuable insight to improve the activity.

■ PRESENTATION OF THE ACTIVITY

"Ethics against Chemistry" was inspired by the popular board game "Clue", in which the participants play the role of investigators trying to solve a murder. To the best of our knowledge, this is a fresh form of science dissemination activity that unites real-life storytelling and gamification, while conveying historical, ethical, and chemistry concepts. In the Supporting Information of this article, we present guidelines for this activity: the plot and rules of the game. By doing so, any investigator will be able to reproduce (and adapt) this activity with other students.

The Game Board

The layout for the game board of "Ethics against Chemistry" is shown in the Supporting Information. The four colors represent the teams and where they put their pins, and the "Start" section marks the beginning of the game. The game board serves as an anchor for the students, marking the position of the teams within the story, revealing the stage of the game and how many stations are left.

Although the literature presents several game frameworks identifying an educational activity as a game, in this case, the definition adapted from Henricks's work is appropriate.[49] The key characteristics of a game as a scientific activity are

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- Playing is a learning experience, as students face new concepts while participating
- Playing a science-based game has intrinsic motives, such as exploration and curiosity for the scientific experiments
- The process of playing the game (e.g., solving the game stations) is more educationally important than the outcome
- Playing a science-based game encourages active engagement from the students as they are the ones who conduct the embedded game experiences)

Introducing the Game and Its Context

A group of 16−20 students checks in to take part in the activity, scheduled to last a full day. Students are grouped into four teams and assigned a team color. The activity uses a game board and a small pre-prepared workbench for the experiments in the classroom. Teachers introduce some details of the story (but do not reveal that it is based on a real-life case) and the rules of the game. The script is based on historical records of the real-life murder case "Crime of Flores Street", historically known in 19th century Portugal as the "Urbino de Freitas Case".[50] In short, a renowned medical doctor was accused of poisoning his nephews to attain the rights to the family heritage. At the time, it was a major criminal case that paved the way for use of modern forensic chemistry in the judicial system.[50]

Whenever the teams move from one station to another or a laboratory experiment is presented (as we will see shortly), the transition is marked in the classroom by playing an animation. These video segments introduce additional elements to the game's story, such as character actions, including some tips for solving each station.

Station 1, "Casa Sampaio"

The teams move to the first station in the game board, where students learn the story of the murder from the characters through a digital animation presentation. The station is named after the family house where the crime was committed. The students' work groups analyze the story and write down possible suspects and their *modus operandi* on their digital game report. Students are encouraged to examine all the details carefully, not only the dialogue of the characters but also the scenery of the animations.

Character Cards

These are stations in the midst of the plot, where game cards are laid on the game board, one for each character who is introduced in the game. Teams analyze the cards, containing an image and a small description of the character who is going to be interviewed by the investigator. Then, students watch a digital animation with the characters' statements. This makes the game board as an anchor setting, stressing the elements and characters the students need to pay attention to. The cards are made available for consultation throughout the game, if requested.

Stations 2 and 3, "The Laboratory"

After the teams meet all of the characters in the Station 1, Casa Sampaio, they move to the Station 2 and Station 3, where students are presented with the pieces of evidence gathered at the crime scene, knowing that at least one of them was used to murder the victim. The presented samples are hydrogen peroxide, a hair sample, copper sulfate, ascorbic acid, an alkaloid, and borax solution. The hair sample is not introduced as a "murder weapon" but to address that the hair might belong to a suspect. Each team will analyze all of the samples for identification, using the pre-prepared laboratory stations. The laboratory procedures are written so the students can execute the experiments, always accompanied by a teacher. All conclusions drawn from the experiments and writing in the game report are the findings of the students. The purpose is to analyze the presented samples, following experimental procedures available in the laboratory, to identify the poisonous substance. Per game rules, to advance to the next station of the game board, all teams must conduct the experiments and write down their findings.

For logistic reasons, the teams take turns to use the laboratory stations. For each experiment, there is a basic chemistry issue (explained after each experiment) or a moral/ethical issue to be addressed. The laboratory stations are divided to accommodate other program activity schedules (such as lunch breaks) and to facilitate logistics so most groups could do the same experiment simultaneously. A summary of the laboratory activities can be found in Figure 2.

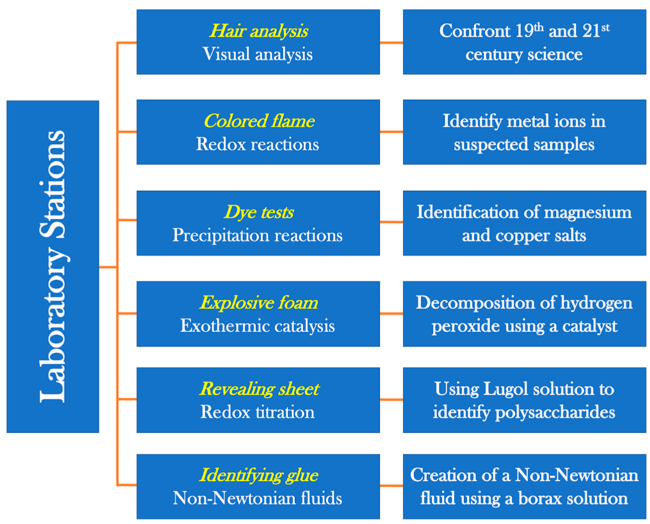

Figure 2. Description of the laboratory hands-on activities (on the left) and its goals (on the right).

Experiment 1: Students Analyze Hair Samples. The purpose is to compare contemporary analytical methods with techniques used in the 19th century. Students are provided with magnifying tools to study the hairs and compare them with images of ampliations of different hair types. They should conclude that it was not possible to assess blame back then using only a hair, as DNA forensic techniques were not available. This gives students the perspective of the evolution of scientific knowledge and techniques, assessing the ethical implications of blaming someone only by circumstance.

Experiment 2: Students Execute a Simple Flame Test and Identify Copper Sulfate. They apply a flame to the solid substances and verify whether the color of the flame changes. By comparing the flame color with a library of photos, they can identify the presence of a metal ion in two samples, enabling them to detect the presence of two salts. Afterward, it is explained to the students the oxidation−reduction process that occurred. This experiment identifies copper sulfate.

Experiment 3: Students Use a Dye Test to Identify Alkaloids. Through a simple dye test, using a commercial Marquis reagent, students apply small quantities of the solid

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samples in an analysis well. After a short period, they check to see whether the reagent changed its color, forming a characteristic precipitate upon interacting with the substance. The experiment finishes with a comparison with a print table describing the toxicity of several substances.

Experiment 4: Students Identify the Presence of Hydrogen Peroxide. The teams are presented with the concept of catalysis with an experiment using the decomposition of hydrogen peroxide. By adding samples in selected vials containing concentrated hydrogen peroxide, they check whether an exothermic reaction occurs, as foretold by the formation of "warm foam". Though the name of the experiment is called "explosive foam", no explosion occurs.

Experiment 5: Students Interact with Redox Reactions, Using the "Revealing Sheet" Experience to Identify Ascorbic Acid. By tainting paper sheets with a provided iodine solution, the groups place small amounts of solid evidence samples and observe what happens. If the brown-colored sheet turns back to its previous color at the place of contact with the sample, it reveals that a redox reaction has occurred. It involves iodine species (iodide ion, free iodine, or iodate ion) and redox reagents in the presence of starch.

Experiment 6: Students Identify a Borax Solution. Students will address the "identifying glue" experiment. Using liquid samples, they will explore the notion of non-Newtonian mixtures by adding volumes of samples to a mixture containing a blend of water and glue. The formation of a specific solid provides the students with a sensory (palpable) experience that the students enjoyed.

By completing the experiments, the teams will identify one of the samples as an "alkaloid", a pivotal element in the investigation and the chemical used to poison the victims (Experiment 3). The remaining experiments were also linked with some details of the storytelling. For instance, the sample of vitamin C could have come from the fruit salts given to the grandchildren (see the Supporting Information). The experiments were designed to emulate a scientific procedure, obligating students to investigate their hypothesis and plan a conclusion. The entertaining factor was also important in experimental design and exemplifying different chemistry topics, mentioned in seventh and eighth grades in Portugal.

Between the two laboratory stations, two characters appear in the animation—Augusto António Rocha and António da Silva—each one representing a laboratory of the game board. Augusto António Rocha insults António da Silva, stating that António da Silva is a liar. Furthermore, Augusto António Rocha confesses he studied medicine with Vicente de Freitas and will do anything to clear his name. This situation generates a conflict of interests, leaving students to discuss which results should be considered, as an example of ethical implications in a scientific problem. They should conclude that Augusto Rocha's statement can be influenced by his friendship with a possible culprit.

Station 4, "The Courthouse"

This is the "dramatic disclosure" station. The trial of this case occurs here, as students use the data gathered from the experiments to discuss their findings. The teams are encouraged to debate in an orderly way, making arguments on who they think is the culprit. For better time management on eventual replication attempts of this activity, this station will be adapted according to the number of students to not spend too much time on giving every team a chance to speak. In the end, before the teams submit their verdict, the fictional prosecutor also makes some arguments in the animation, which introduces an ethical issue. The students find out that the prosecutor had a previous romantic affair with a wife of a suspect, generating another conflict of interest. Then, students discuss if this statement should be considered or not and if it would change their suspicions on who is the perpetrator. After that, each team submits its digital reports containing their definitive answer.

Station 5, "Academy of Porto"

The last station is the "revelation station". The students are presented with a scientific conference-type setting (hence the name "Academy of Porto"), where the animation presents one eminent scientist that, in real life, helped to solve the case: António Ferreira da Silva. He states the *modus operandi* of the murder: alkaloid poisoning (detected in Experiment 3). Also, by linking the laboratory experiments with the story (as who could handle and get such a substance as the alkaloid used), he reveals the culprit as being Vicente de Freitas.

Finally, the teachers announce the teams that successfully determined the culprit and declare the winning team based on the game rules.

■ ACTIVITY IMPLEMENTATION

In July 2019, students from all over Portugal and some European countries signed into a week's program of the University of Porto's "Universidade Júnior", containing different activities every day (including "Ethics against Chemistry"). Students freely choose among the activities of the program what activities in which to participate. This activity was available for 20 sequential working days, each day with a different class, divided randomly into four teams of 3 or 4 students each. In total, about 280 students were distributed into 80 teams.

Knowledge assessment and activity evaluations were collected using the activities' online webpage, accessed by the students through a smartphone with Internet access (one per team). Because of its versatility, smartphones were chosen as a means to access different platforms used in this activity. Also, we could collect data in real time; as students filled in the game report, we could track each team's performance.

The first platform was the game report, where students wrote their conclusions and suspicions during the game. After completing each task assigned to a specific station in the game board, teams could move forward to the next one. Another platform was an interactive 10 question quiz (with the same teams), which covered various aspects of the game. The last tool was a voluntary individual inquiry to evaluate the activity. The goal of the 10 question quiz was to evaluate whether students were attentive during the activity; the voluntary inquiry was designed to obtain participants' feedback and opinions of the activity. The game report was meant for students to give their opinion on ethical dilemmas that the characters were facing and to fulfill a laboratory report of the experiments.

■ RESULTS

The performance of the students was tested with an empirical assessment of the scientific reports made by each team. The questions on each experiment were evaluated according to the clarity of thought, identification of chemical principles, and their deduction capabilities on their solutions in each activity. The queries evaluated are presented in Figures 3 and 4.

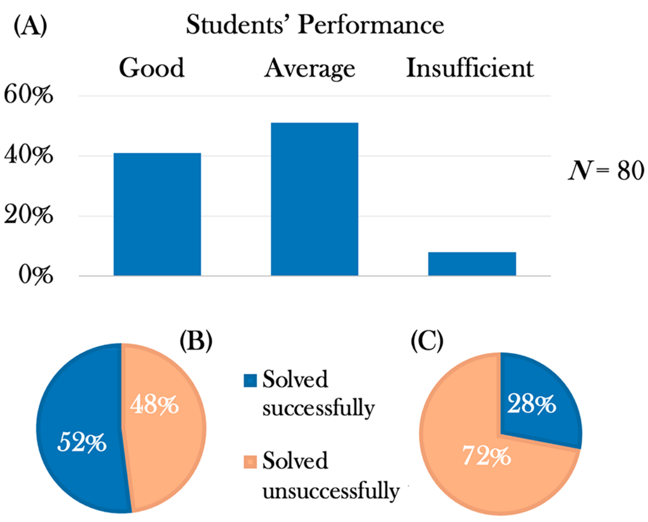

Figure 3. (A) Graphic bar, dividing the students' performance into three levels, N = 80 teams: good, average, and insufficient. (B) Of the teams who rated a good performance, 52% solved the case successfully, whereas 48% did not. (C) Pairing the data from the average and insufficient group, only 28% solved the case successfully, whereas 72% did not.

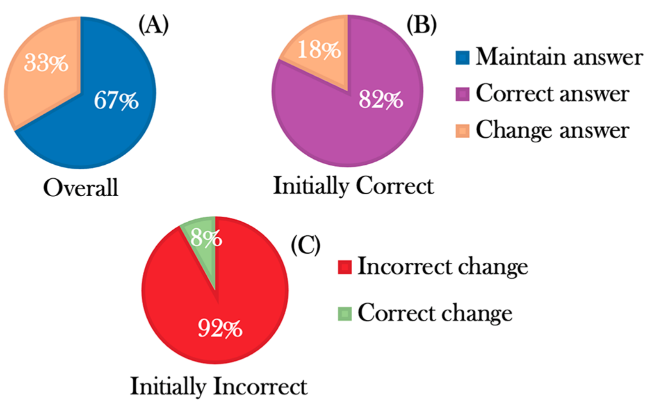

Figure 4. Circular graphics analyzing the students' suspect definitive answer, N = 80 teams. (A) After the ethical dilemma, 33% of teams changed their definitive answer, whereas 67% did not. (B) Of the teams that had the right answer during the game, 18% changed their answer in the last question, whereas 82% maintained the correct option. (C) Of those who changed their answer, 8% gave the correct answer, whereas 92% did not.

Students were grouped into three categories according to their performance, characterized as good, average, and insufficient. This assessment was based on three principles tested on their open-and-closed answers: correctly identifying the composition of the samples and the chemical principles and describe chemical reactions; successfully identifying the murderer and poison used to commit the murder; and showing understanding of ethical implications of scientific procedures. Table 1 presents the criteria for classifying a team's performance. A teams' success is attributed when they can correctly identify all of the samples in the crime scene in the laboratory stations. Also, they need to identify the culprit and what sample/substance is used to commit the crime. For the ethical implications, we stated two situations that could be mentioned: the friendship of Augusto António Rocha with Vicente de Freitas that could lead to producing false results to help him; the question of the interests behind the prosecutor, as he was a previous lover of a suspect's wife. Since each classification has three possible criteria, if a team presents at least 2 out of the 3 presented, they classify on that mark.

Table 1. Activity Evaluation Rubric for "Ethics against Chemistry"

	characterizations rubric by category		
team evaluation parameters	good	average	insufficient
identification of sample compositions	>75%	>50%	<50%
identification of the murderer and poison used	both correct	1 correct	0 correct
recognition of ethical implications	1 or more recognized	none recognized	none recognized

As mentioned before, the last question of the game report (who was the murder and why?) was crucial for team evaluation performance. We selected answers (and translated them) as an example of their performance. Teams that had a good performance answered: "It was Dr. Vicente that poisoned Mário because Dr. Vicente intended to gain the family's money along with Maria das Dores, putting alkaloids in the coconut cake". Teams that had an average performance answered: "It was Maria Luísa that killed Mário because she wanted the money for her son, and she gave the coconut cake to the children". Teams that had an insufficient performance answered: "Maria das Dores because she wanted all the money".

Among the students with good performance, 52% of students solved the crime successfully, whereas 48% did not. However, many students with average or insufficient performance could not solve the problem. There were more students with average or bad performance, and a higher percentage of those students could not solve the problem. Only 28% of the students attained the correct answer, with a substantial percentage of 72% of team members who could not (n = 80).

The reaction of the students upon ethical dilemmas in a scientific context was also evaluated. For example, a controversial issue was brought up, as the final animation suggested the prosecutor had a former romantic relationship with the suspect's wife. Although this did not affect the outcome of the laboratory's stations and the case, many of the students believed, as stated in the digital reports, that this conflict of interest should not be considered as evidence. However, when asked who committed the murder, 33% of the students were persuaded by the prosecutor's relationship and changed their answer, and from these subjects, 92% made an incorrect change, meaning that they were heavily influenced by this outcome. Another interesting element of analysis is the chosen suspect by the teams' changes as they execute the experiences. In our assessment, 82% of students maintained a correct assertion of the culprit along with the game, as only 18% changed at the last stage of the game (n = 80). These students failed to present a viable cause for the suggestion of their suspect, meaning that they were somewhat influenced by the nature of the story revelations made.

The retention of information was also evaluated in this activity. In a final task, students were prompted to answer a 10 question essay regarding various aspects of this activity, from scientific experiments to storytelling elements. The same teams

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took part, and we concluded they could get right roughly about 8.7 answers right out of 10 (n = 72).

The evaluation of this activity was also embedded in the students' digital application. At the end of the activity, students were asked to grade the complete experience giving a 1−5 rank (Likert scale[51]), with 1 being rated as awful and 5 as excellent. The participants gave the activity a 4.77 grade (n = 160).

When asked what they have learned from this activity, the answers obtained can be grouped into the following major categories: history of science; investigation (highlighting the observation skills); chemistry/compound identification; positive attitude toward chemistry or science (referring to its utility and describing as interesting and likable); and social skills (such as teamwork). The assessment of these findings was obtained by word analysis from their report answers.

■ DISCUSSION

An assessment of evaluation based on these results can be divided into three specific vectors of action: students' scientific performance; ethical judgment; and information retention. Only about half of the teams achieved "good" performance. Therefore, one question to address is why most students had an "average" or "insufficient" performance. When comparing the raw data from the students' reports and the empirical statistical value of their performance, a couple of issues surface. Several teams revealed they had no previous laboratory training and therefore could not identify most of the reagents and appliances. Only the ones that had some theoretical basis on the topics addressed in each experiment could easily respond and solve the questions in each station. One of the most common mistakes seen during the laboratory experiments was sampling errors and students forgetting to follow the laboratory safety rules. This shows the low preparation of middle school students when facing chemistry practical activities in which applied previous knowledge is needed.[52] The lack of fundamental answers in the activity reports and considering that they came from different backgrounds and social-economic status can give a glimpse of the ill-preparedness of students to apply school-taught science topics in real-life situations.

One must not forget that this activity was included in a summer program that is supposed to be dynamic and fun for the students, defined as nonformal teaching.[53] There was no real classroom evaluation and no pressure to succeed in each task (besides winning the game), which probably had a vast influence on our results.[54] It might be possible that if this activity happened in the children's classroom with strict evaluation and not inserted in a summer program the children could have achieved better results or at least put more effort into the game report fill-in. Although students were encouraged to write everything down in their digital reports, perhaps some parts of the report should be reconsidered in future editions. This could also help manage the timeline of the activity with a heterogeneous class (with different writing skills), saving time on the report fill-in, and could also avoid boredom in teams since the game could only move forward when every team finishes their digital assessment.

On the other hand, students were frequently paired with someone they just met. This factor can influence the students' participation within group discussions and, ultimately, the final answer on the digital report. Future iterations of this activity would benefit if the students are already familiar with each other.

The final ethical question posed to the students identified some issues regarding their process of decision-making while playing the game. As reported, most teams could combine the information provided in each station methodically, allowing them to write down the correct suspect. Since 82% of the students maintained their right assumption, these teams showed they could separate the science facts from the conflict of interests provided by the characters. These results open the path to new forms to discuss conflicts of interest in science as it can, and should, be done with young students.

However, some teams were heavily influenced by this ethical issue that had no consequence on the outcome of the case. When asked about the hypothetical influence of a third party, they were eager to attribute the answer toward this entity, disregarding all the scientific evidence they amassed earlier. In this evaluation, about one-third of the students changed their answers when addressing this issue. Within this group, 92% of them changed to an incorrect answer. Therefore, we can understand that a small part of middle schoolers was easily influenced by drama when ethical issues were mixed up with scientific discernment.

Our major challenge during this activity was directing student focus as the story is being told. They easily became distracted and created new and complex hypothesis to explain the murder that were not coherent with the narrative. These include attributing the murdered to an outside character or even questioning the murdered itself, hypothesizing a potential suicide.

On the matter of information retention, in the 10 question survey, the students outperformed with an average of 8.7/10 correct answers. Since the questions were regarding the scientific procedures performed in the activity and details of the story, one may conclude that they were attentive during the activity and were able to preserve most of the details asked. Although this result seems to contradict the previous results mentioned in Figure 4, the type of inquiry was different, evaluating detail retention, such as the year of this case. On the other hand, results in Figure 4 evaluate whether teams responded correctly as the murderer being Vicente Urbino de Freitas. If teams failed to answer correctly, one cannot extrapolate that they ignored the activity. They were just not able to connect the information on the story with the results of the laboratory experiments. To resolve this issue, the digital animation should be altered to include the question: "Who could obtain the poison (alkaloids)?" By doing so, it could lead the students to reflect on who, in the 19th century, could have access to it. A medical doctor with colleagues working in a laboratory could have easier access to such substances.

The results of the activity assessment show that the participants enjoyed it, were curious, and had a great recognition of scientific elements presented within the story. This shows the potential landmark for establishing a gamification-based storytelling activity toward the dissemination of scientific topics.

Although these are promising results, further investigation is still necessary, as there were other investigation outlines we did not explore. Long-term information retention of the participants, an evaluation of previous chemistry knowledge, combined with an after-activity assessment, is also relevant. The first year of implementation of this activity was merely exploratory, with a solid purpose to put "Ethics against Chemistry" to a test using the feedback from the students to improve it. One of the main proposals on how this game could

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be improved should address that most of the students were underprepared and some of them could not solve the crime successfully. Roughly, this could be mitigated with a prior introduction of the laboratory part of the activity to the students (giving more time to understand what they are required to do). Also, evaluation methods still require some improvement to convey more outtakes from the gathered data.

While we followed a script and a timeline, different situations occurred every day (some classes were more engaged in the discussions than others), as we had to adapt the activity to each heterogeneous class. Standardization of the activity is required to test it with a larger sample to obtain more robust results. Also, future editions should be done with classes with similar levels of chemistry knowledge. This way, teachers can better prepare the activity and avoid boredom of students with higher levels of chemistry education. Future editions should include better time management for discussion, as often teams were extremely participative, making it difficult for every student to pitch their suspicions.

The results also reveal a positive impact on other adaptations and reproducible efforts of this activity. This exploratory experience provided qualitative data that allow several structural and methodological improvements, which with the materials provided in the Supporting Information allows any investigator to implement and improve this activity.

"Ethics against Chemistry" is a refreshing form of edutainment that uses for the first time a gamification-based storytelling foundation to disseminate scientific and ethical topics. The application of real-life facts from the history of science, embedded in a storytelling scenery, proved to be successful in addressing science themes for middle school students. The game-oriented participants to execute several chemistry experiments that amassed scientific facts toward the solution for the outcome of the crime they were trying to solve. This outtake was well received by students, as many of them performed fairly when addressing these issues in a laboratory settlement. Using this activity as a benchmark for future activities, "Ethics against Chemistry" aims to be a beacon for scientific events that promote analytical thinking, allocating the entertainment factor, well suited for middle school students.

■ CONCLUSION

Based on our results and the feedback given by students, we can say it was a successful first edition of "Ethics against Chemistry". The students' enthusiasm and curiosity give us reasons to uphold that this edutainment activity established a great learning framework, even for difficult or unattractive subjects, such as chemistry and ethics.

The model of this game, or a similar activity, can be a good starting point to discuss other scientific topics and history of science episodes in a nonformal educational context. We provide a basic blueprint for a similar activity, aiming to address basic chemistry educational purposes. Although with the appropriate adaptations, "Ethics against Chemistry" could be an excellent resource to be used in science classrooms of the seventh or eighth grades.

Using a real-life event from the history of science and the location of this event (Porto, Portugal) contributed to an easier connection with the audience (Portuguese students). Also, it allowed them to assimilate the importance of the history of science, its overall development, and how it can explain much of science applications in our daily life. Therefore, it was a major contributor to increase students' awareness of science during this event.

Further studies are needed to evaluate other important parameters to assess the full scope of this approach. Studies on motivation, awareness of science, and long-term knowledge retention are needed to clarify how gamification can help children's learning patterns, attending middle school science classes.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available at https://pubs.acs.org/doi/10.1021/acs.jchemed.0c01469.

> Timeline of the activity; script and game elements used in "Ethics against Chemistry" (storyline, script synopsis, cards, game board, still shots from the animation); knowledge assessment questions (with correct answers indicated); laboratory manual with instructions for the laboratory experiments (hair analysis, colorful flame, dye test, "explosive" foam, identifying glue, revealing sheet), and all safety and hazard information for implementing the activity (DOCX, PDF)

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Notes

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